SAKSHAM UPADHYAY

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Education

VIT Bhopal University — Bhopal, MP
Bachelor of Technology — Cumulative: 8.36/10
Major in Electronics and Communication Engineering — Graduating in July 2026
Relevant Coursework : OOPS, DSA, DBMS, System Design(HL)

Technical Skills

Languages: Java, Python, HTML, SQL

Libraries/Frameworks: NumPy, Pandas, Scikit-Learn, TensorFlow, Keras, Tailwind CSS, LangChain, CrewAI

AI/ML Expertise: Machine Learning, Generative AI, Natural Language Processing, Deep Learning

Databases: MySQL

Projects

CodeCritic AI | Python, LangChain, Google Gemini API — July 2025 – August 2025

- Automated code analysis for **10+ programming languages**, reducing manual review time by **80%**.
- Designed **dual-mode LLM engine**, combining rapid feedback and deep multi-tool analysis.
- Reduced developer workload by **15 hours per week** with automated categorization into **6 priority sections**.
- Improved cost-efficiency by **25%** via caching and error handling across **5+ modules**.

DocGenie | Python, LangChain, Google Gemini API — February 2025 – March 2025

- Built a **multilingual AI PDF chatbot** capable of answering queries from **100+ page documents in 4+ languages**.
- Integrated **semantic search with FAISS** over **1,000+ text chunks**, boosting retrieval accuracy by **30%**.
- Added **voice-based query handling** (gTTS), improving accessibility for **100% of users**.
- Designed a Streamlit UI supporting **10+ user sessions** with persistent chat history and export options.

Deep Fake Audio Detection | Python, Deep Learning, Audio Processing — November 2024 – December 2024

- Constructed a CRNN model achieving 73%+ accuracy in detecting audio deepfakes with 3 convolutional layers and a 2-layer bidirectional LSTM, trained on 1,000+ real/fake audio samples.
- Optimized audio preprocessing by standardizing variable-length inputs to fixed 200-frame spectrograms (16 kHz SR, 1024 n FFT), reducing inference latency by 30% through GPU-accelerated batch processing (16 samples/batch).
- Reduced overfitting by 15% via dropout (0.3) and adaptive learning rate scheduling (patience=2), while maintaining 90%+ validation accuracy across 10 training epochs.

Certifications

Generative AI Certification - IBM (April 2025)

HTML, CSS, and Javascript for Web Developers - Coursera (December 2023)

Java Programming Certification - SkillVertex (March 2023)

Extracurricular Activities

- **Core Member, Linpack Club (PR Outreach Team):** Led outreach campaigns, growing club visibility by **40%** across campus.
- **Core Member, Edu4U Club (Event Management Team):** Organized **5+ events** with **200+ student participants.**